Linux File Permissions

Project Overview

This hands-on project demonstrates how to inspect and modify Linux file and directory permissions. I used command-line tools to review existing permissions, apply selected changes with chmod, and verify the results with ls -la. The exercise focuses on access control for file owners, groups, and other users.

1. Check the current location

I used the **pwd** command to display my current working directory. This confirmed that I was working in **/home/researcher2** before navigating to the projects directory.

```
researcher2@0659a74f1e34:~$ pwd
/home/researcher2
researcher2@0659a74f1e34:~$ cd projects
researcher2@0659a74f1e34:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 17:33 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 18:13 ..
-rw--w---- 1 researcher2 research_team   46 Oct 14 17:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 17:33 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Oct 14 17:33 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 17:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 17:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 17:33 project_t.txt
researcher2@0659a74f1e34:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@0659a74f1e34:~/projects$ ls -la
```

Next, I changed into the projects directory and ran **ls -la**. This command lists files and directories, including hidden files, and displays their permission details. The output included **project_x.txt, project_k.txt, project_m.txt, project_r.txt, project_t.txt, the hidden file .project_x.txt, and the drafts directory.**

2. Understand the permissions string

Linux displays file and directory permissions as a ten-character string. For example:

-rw-rw-r--

The first character identifies the item type: '-' indicates a regular file and 'd' indicates a directory. The remaining nine characters show permissions for the owner, group, and others.

- • r — Read permission
- • w — Write permission
- • x — Execute permission
- • - — Permission not granted

I used ls -la to inspect these permission strings and identify which permissions were granted to the owner, group, and others.

3. Change permissions on regular files

I removed group write permission from project_r.txt and project_t.txt using the chmod command:

chmod g-w project_r.txt
chmod g-w project_t.txt
ls -la

The g-w mode removes write permission from the group. It does not remove write permission from others. I ran ls -la again to verify the updated permissions.

The updated output showed that the group write bit had been removed from both files.

```
researcher2@0659a74f1e34:~/projects$ chmod g-w project_r.txt
researcher2@0659a74f1e34:~/projects$ chmod g-w project_t.txt
researcher2@0659a74f1e34:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 17:33 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 18:13 ..
-r-------- 1 researcher2 research_team   46 Oct 14 17:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 17:33 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 17:33 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 17:33 project_m.txt
-rw-r--r-- 1 researcher2 research_team   46 Oct 14 17:33 project_r.txt
-rw-r--r-- 1 researcher2 research_team   46 Oct 14 17:33 project_t.txt
researcher2@0659a74f1e34:~/projects$ cd drafts
researcher2@0659a74f1e34:~/projects/drafts$ ls -la
total 8
drwx--x--- 2 researcher2 research_team 4096 Oct 14 17:33 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 17:33 ..
```

4. Change permissions on a hidden file

Linux hidden files begin with a dot (.). In the projects directory, I worked with the hidden file **.project_x.txt** and ran:

chmod u-w,g-w .project_x.txt
ls -la

The **u-w,g-w** mode removes write permission from the file owner and group. I then used ls -la to inspect the resulting permissions.

```
researcher2@0659a74f1e34:~/projects$ chmod u-w,g-w .project_x.txt
researcher2@0659a74f1e34:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 17:33 .
drwxr-xr-x 3 researcher2 research_team 4096 Oct 14 18:13 ..
-r-------- 1 researcher2 research_team   46 Oct 14 17:33 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Oct 14 17:33 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Oct 14 17:33 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Oct 14 17:33 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 17:33 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Oct 14 17:33 project_t.txt
```

5. Change directory permissions

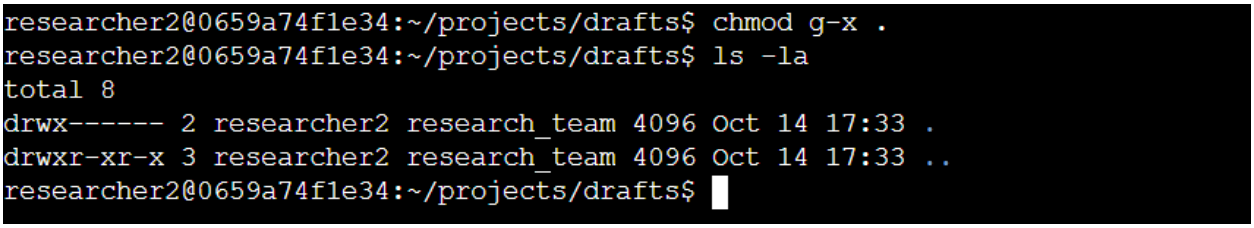
I navigated into the drafts directory using cd drafts.

I then ran the following command:

chmod g-x .

The dot (.) represents the current directory. The command **chmod g-x .** removes execute permission from the group for the **drafts** directory. On directories, execute permission allows traversal and access to entries when the other required permissions are present.

I used **ls -la** to inspect the directory permissions after the change. The output showed that the group no longer had execute permission on the **drafts** directory. This demonstrates permission behavior; it is not necessarily a recommended setting for a shared working directory.



6. Summary

In this project, I practiced inspecting and changing Linux file and directory permissions using **pwd, cd, ls -la, and chmod.** I worked with regular files, a hidden file, and a directory, then checked the resulting permissions. The exercise strengthened my understanding of Linux access control and the importance of verifying permission changes.